

# **CONNECT WITH PLYMOUTH UNIVERSITY**

**High speed communications**

**MIMO Systems**

WITH  
PLYMOUTH  
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# What is MIMO?

- Terminology
  - SISO - Single Input Single Output



SISO - Single Input Single Output

- SIMO - Single Input Multiple output



SIMO - Single Input Multiple Output

# What is MIMO

*Refers to antenna  
outputs and inputs.*

- Terminology

- MISO - Multiple Input Single Output



MISO - Multiple Input Single Output

- MIMO - Multiple Input multiple Output



MIMO - Multiple Input Multiple Output

# How MIMO works

- MIMO takes advantage of multi-path.
- MIMO uses multiple antennas to send multiple parallel signals (from transmitter).
- In an urban environment, these signals will bounce off trees, buildings, etc. and continue on their way to their destination (the receiver) but in different directions.
- “Multi-path” occurs when the different signals arrive at the receiver at various times.
- With MIMO, the receiving end uses an algorithm or special signal processing to sort out the multiple signals to produce one signal that has the originally transmitted data.



# MIMO and Shannon's Law

Shannon's Law states that:

$$C = B \cdot \log_2(1 + S/N)$$

Where B = bandwidth and S/N is the signal to noise ratio and C is the bit rate.

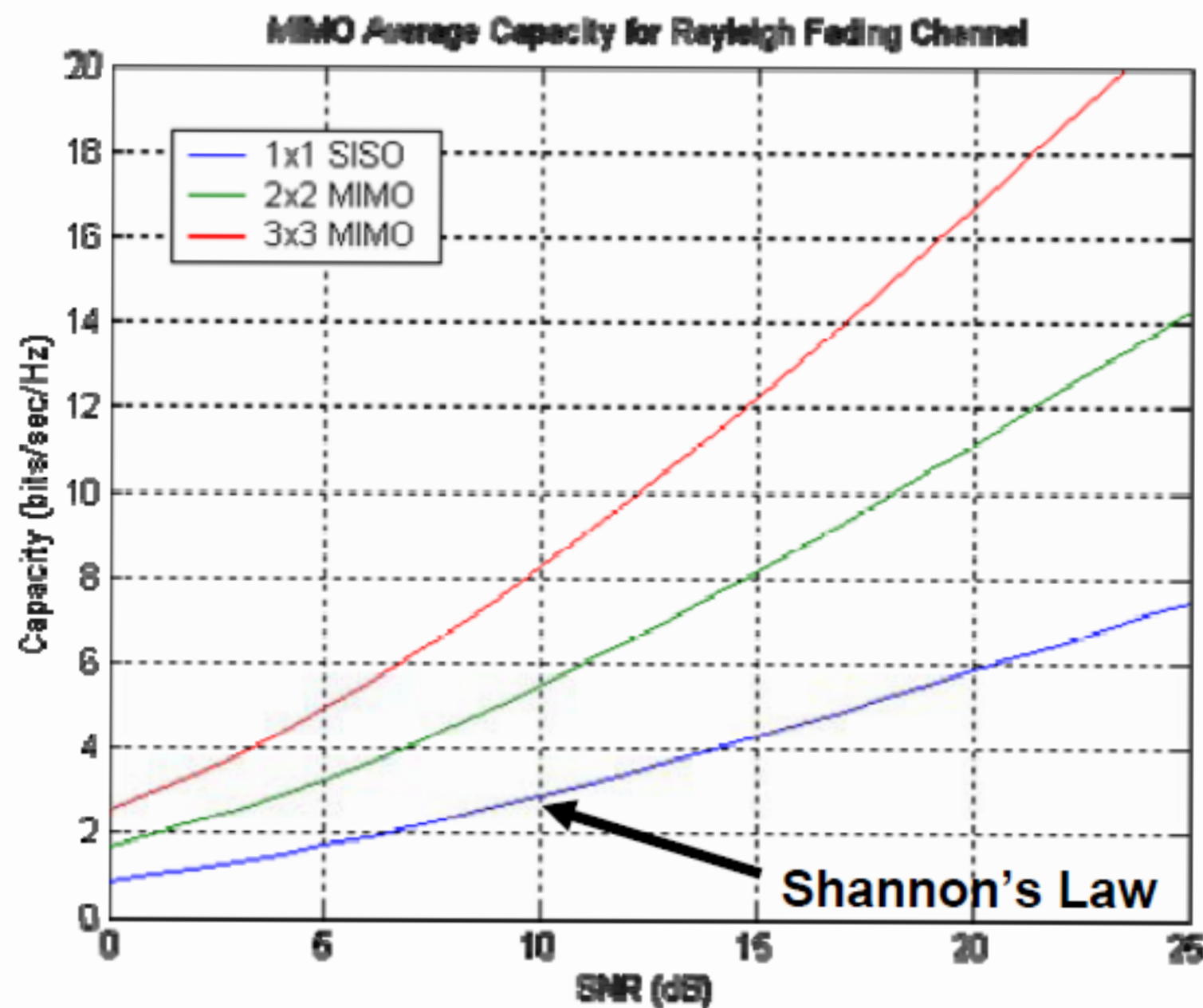
Example:

If the SNR is 20 dB, and the bandwidth available is 4 kHz, which is appropriate for voice communications, then  $C = 4 \log_2(1 + 100) = 4 \log_2(101) = 26.63$  kbit/s. Note that the value of S/N = 100 is equivalent to the SNR of 20 dB.

Shannon Hartley  $\sim 2B \cdot \log_2(M) = B \cdot \log_2(1 + S/N)$  meaning  $M = \sqrt{1 + \frac{S}{N}}$

Bandwidth, SNR, Modulation/signalling levels.

# MIMO increases throughput



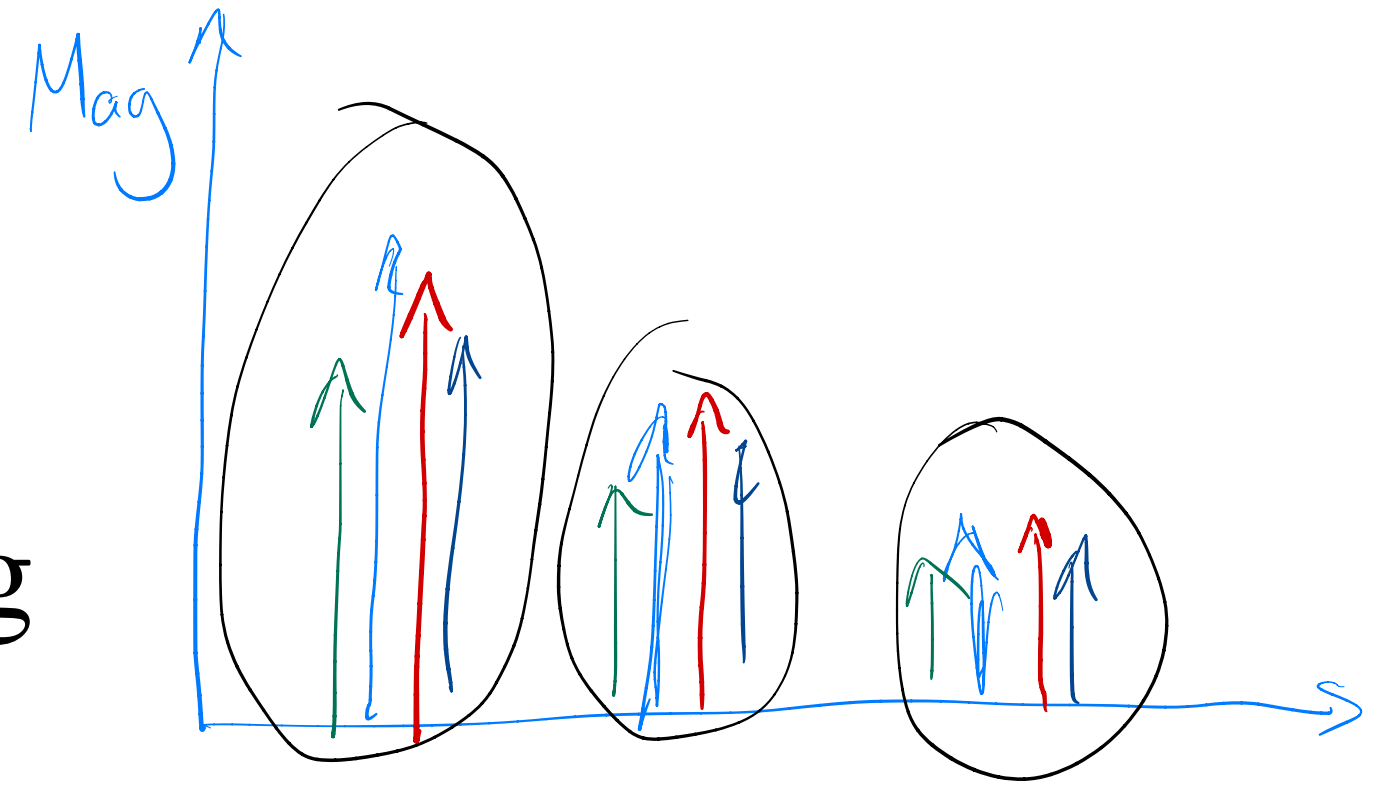
Wireless throughput scales as more radio transmissions are added onto the same channel

Only baseband complexity, die size/cost, and power consumption limits the number of simultaneous transmissions (assuming good channel conditions)



# Spatial Multiplexing

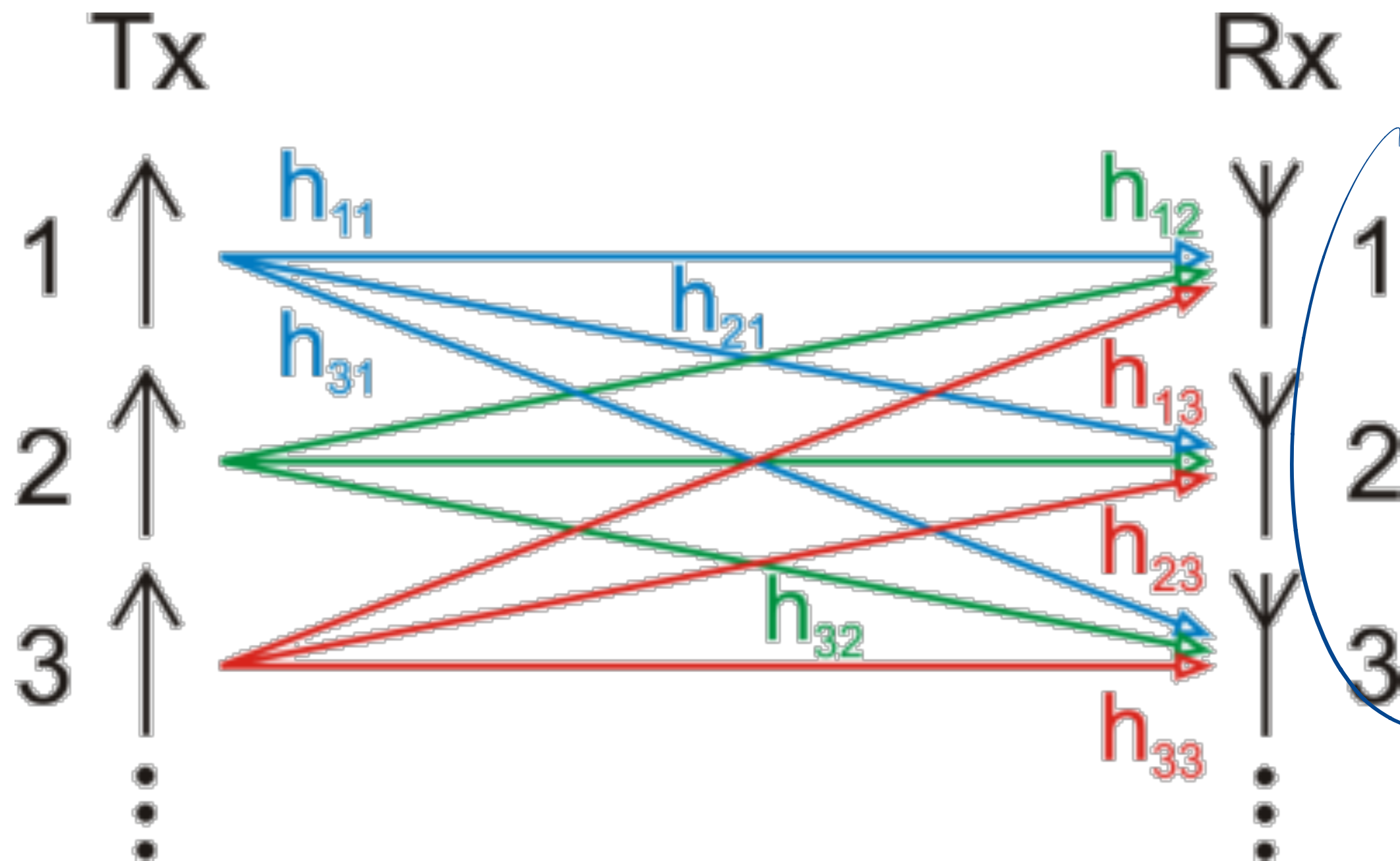
- Phased arrays/adaptive antenna
  - Allow beam steering/forming
  - Gain DOA information
  - Can support SDMA
- MIMO, one step further
  - Similar to SDMA
  - But identifies spatial channels rather than users
  - Does this by using multiple antennas



Multiple antenna allow  
signals to be resolved  
& grouped

# MIMO Spatial Multiplexing

Transmit paths for a 3\*3 MIMO system.



I can group the coloured paths together based on arrival times



# MIMO Spatial multiplexing

1. In MIMO spatial multiplexing the number of receive antennas must be equal to or greater than the number of transmit antennas
2. The capacity increase is related to the number of antenna in a MIMO system. Where  $M \times M$  system and  $M \geq 2$ , increase =  $M$ .
3. The number of spatial paths available is also related to the number of antenna,  $M^2$

$\text{Increase} = M \times M = M^2$  or  $\begin{matrix} 2 \times 2 \\ 2 \times 3 \end{matrix} \Rightarrow \begin{matrix} 2 \times \text{speed} \\ \text{smaller of two } Ms \end{matrix}$   
 $M \times N > M > N = N \text{ increase}$

Previous slide  $3 \times 3 \Rightarrow 9 \text{ paths}$

Receive Channel Trans

$$[R] = [H]^* [T] \text{ i.e. } \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix} \times \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}$$

location means each path (channel) is slightly different

# MIMO Spatial multiplexing

So what is received in our 3\*3 system?

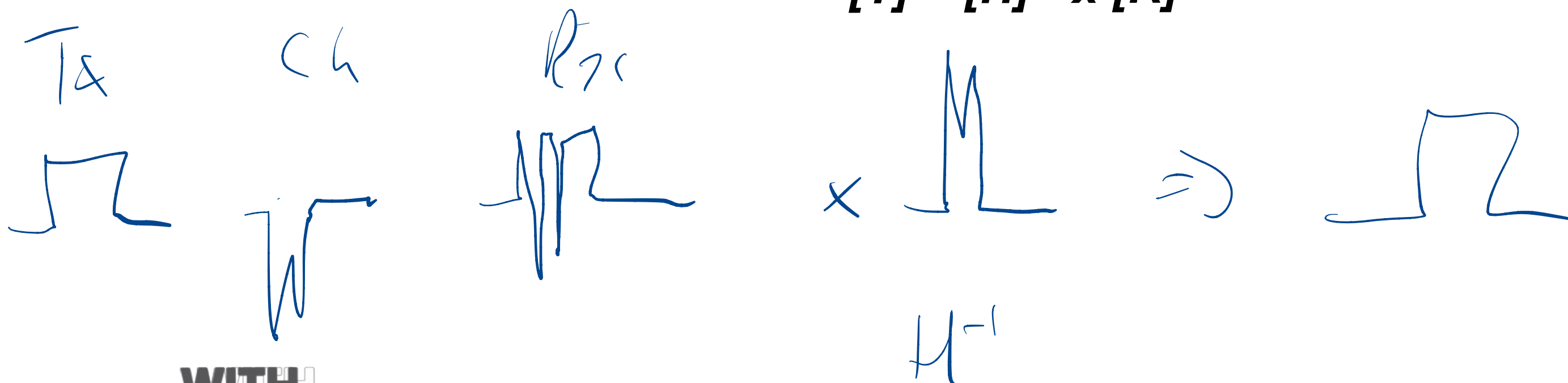
$$r1 = h11 t1 + h21 t2 + h31 t3$$

$$r2 = h12 t1 + h22 t2 + h32 t3$$

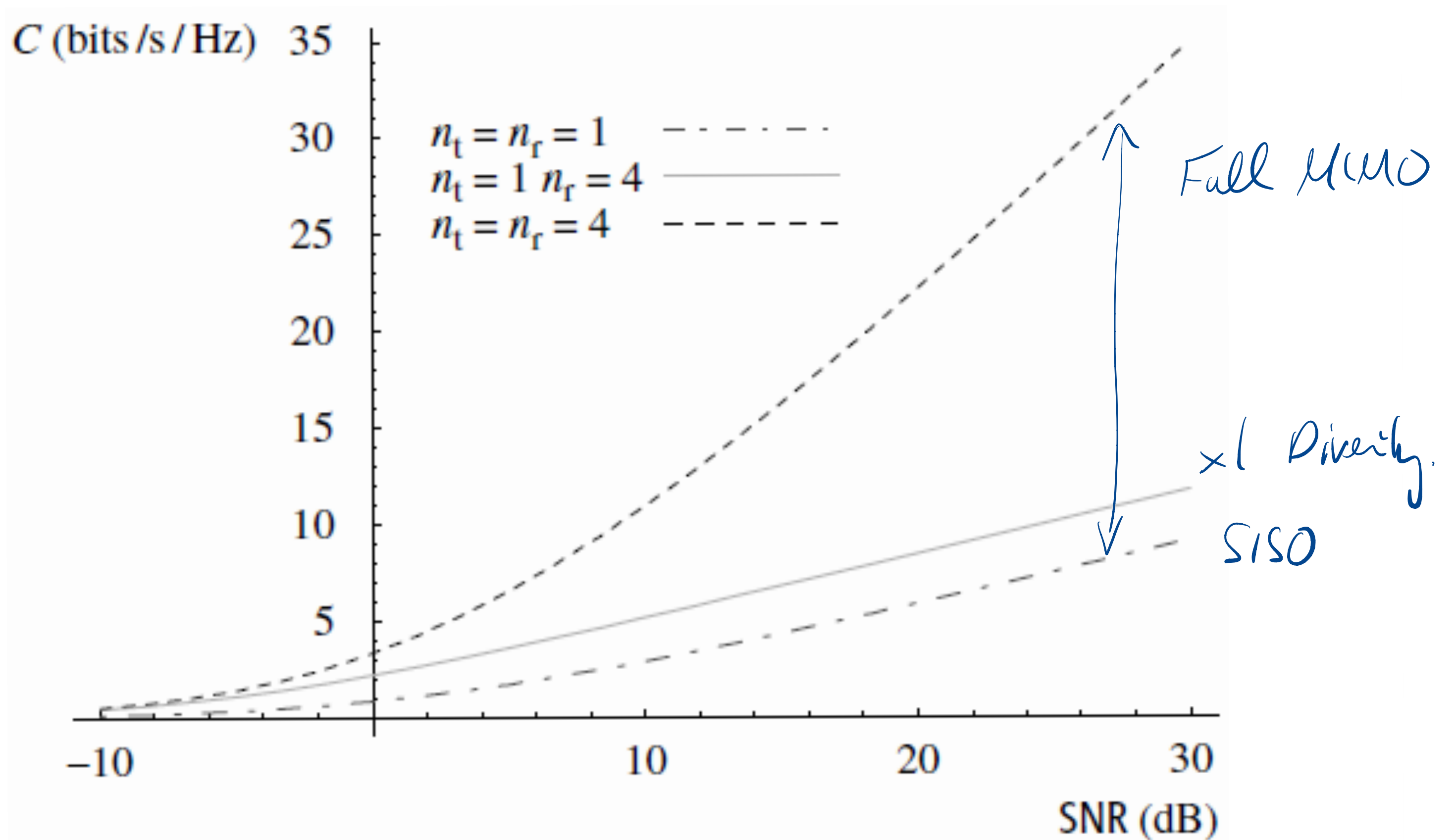
$$r3 = h13 t1 + h23 t2 + h33 t3$$

How do we get our transmitted information back?

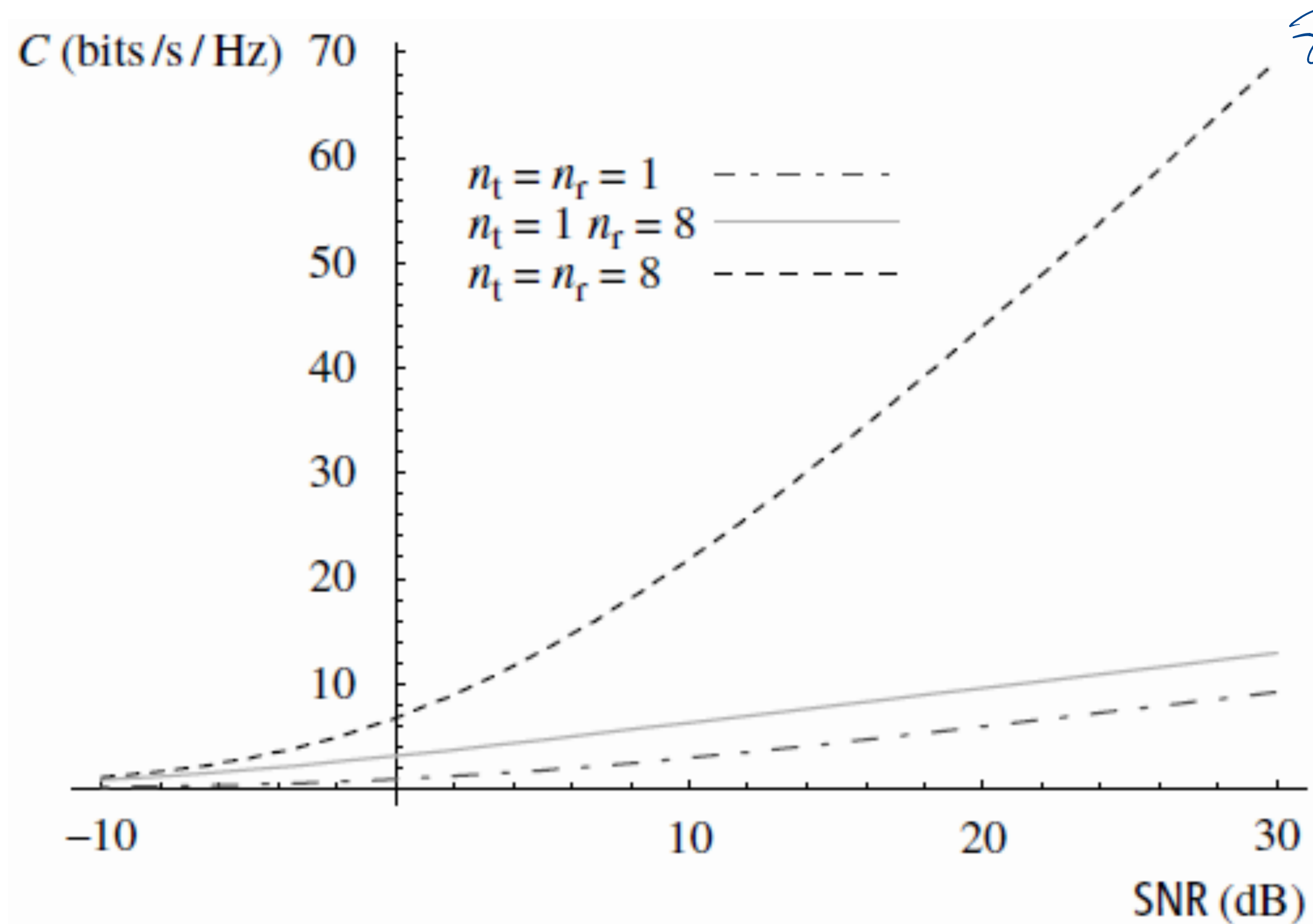
$$[T] = [H]^{-1} \times [R]$$



# MIMO Gains 4\*4 channel



# MIMO Gains 8\*8 Channel



8x8

1x8

1x1 250



# Constraints of the channel

- Since the transmit vector is projected onto the channel matrix  $\mathbf{H}()$ , the number of independent data streams that can be supported is limited by the rank of  $\mathbf{H}()$
- The properties of  $\mathbf{H}()$  determine the potential performance for a MIMO system

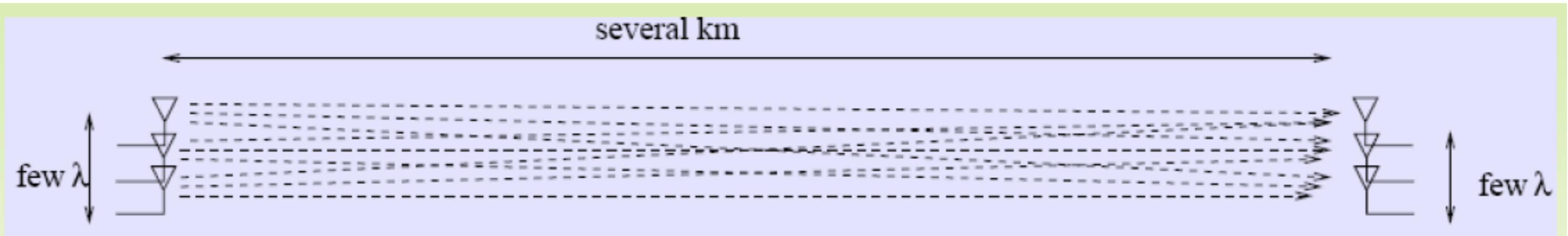
Needs Multipath

Spatial channels need to be different enough to resolve,



# MIMO Spatial multiplexing, strong line of sight

*MIMO Needs Multipath*

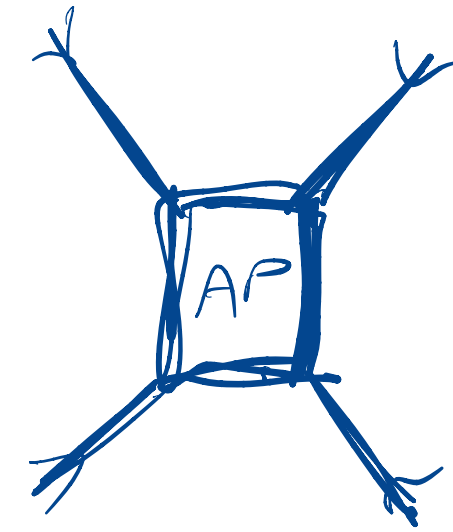


$$\mathbf{H} \approx \alpha \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

The system is near rank one (non invertible)!

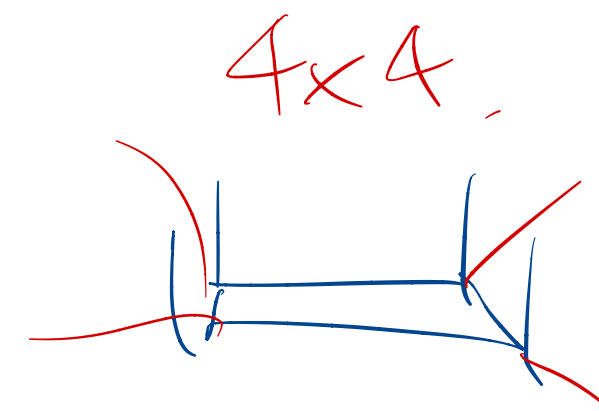
**Spatial multiplexing requires multipath to work!!!**

# The Channel Matrix

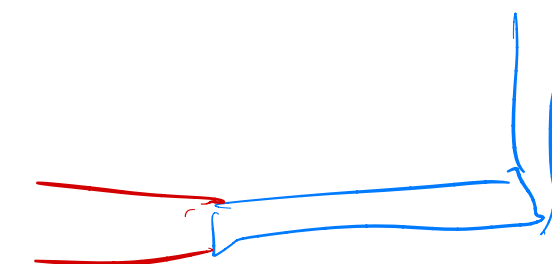


- The channel matrix  $\mathbf{H}()$  comprises the effects of:
  - Array size
  - Antenna configuration
  - Element pattern
  - Element polarisation
  - Element coupling
  - Antenna impedance matching
  - Multipath propagation characteristics
- - no multipath, no MIMO

Which best



Same?



2x2

2x2

Polarized?

# Modulation Bandwidth MIMO

Rate adaptation,

Choosing the best

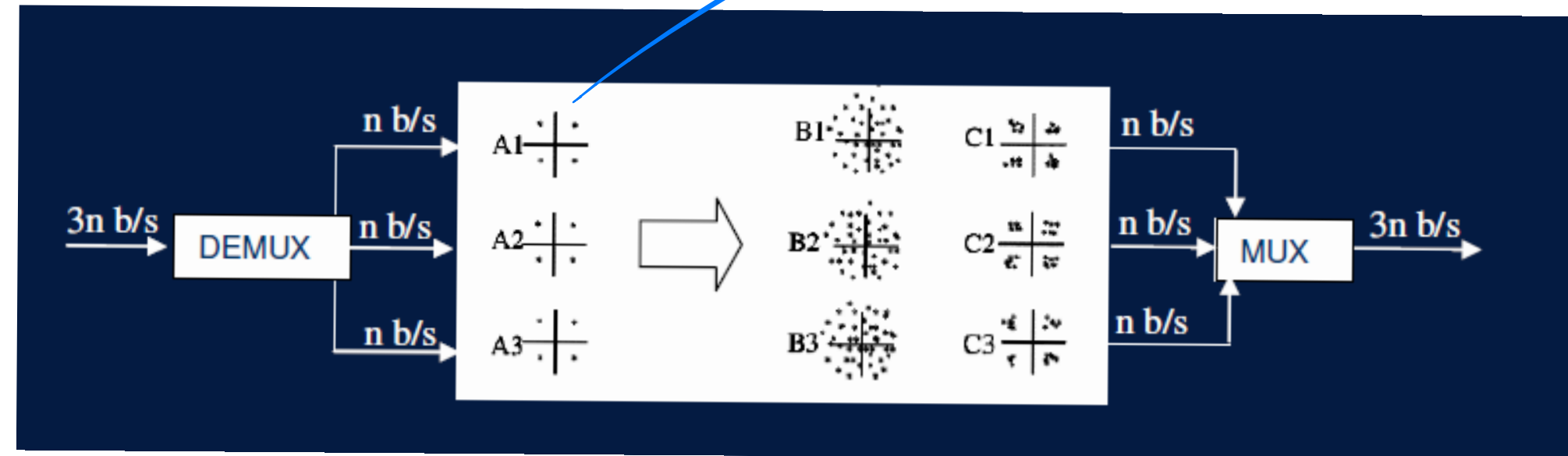
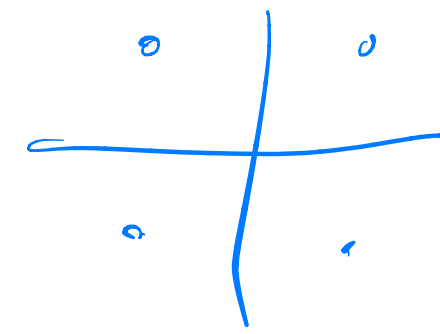
Modulation  
Coding  
Diversity  
MIMO

} For channel.

| MCS index | Spatial streams | Modulation type | Coding rate | Data rate (Mbit/s) |           |                |           |
|-----------|-----------------|-----------------|-------------|--------------------|-----------|----------------|-----------|
|           |                 |                 |             | 20 MHz channel     |           | 40 MHz channel |           |
|           |                 |                 |             | 800 ns GI          | 400 ns GI | 800 ns GI      | 400 ns GI |
| 0         | 1               | BPSK            | 1/2         | 6.50               | 7.20      | 13.50          | 15.00     |
| 1         | 1               | QPSK            | 1/2         | 13.00              | 14.40     | 27.00          | 30.00     |
| 2         | 1               | QPSK            | 3/4         | 19.50              | 21.70     | 40.50          | 45.00     |
| 3         | 1               | 16-QAM          | 1/2         | 26.00              | 28.90     | 54.00          | 60.00     |
| 4         | 1               | 16-QAM          | 3/4         | 39.00              | 43.30     | 81.00          | 90.00     |
| 5         | 1               | 64-QAM          | 2/3         | 52.00              | 57.80     | 108.00         | 120.00    |
| 6         | 1               | 64-QAM          | 3/4         | 58.50              | 65.00     | 121.50         | 135.00    |
| 7         | 1               | 64-QAM          | 5/6         | 65.00              | 72.20     | 135.00         | 150.00    |
| 8         | 2               | BPSK            | 1/2         | 13.00              | 14.40     | 27.00          | 30.00     |
| 9         | 2               | QPSK            | 1/2         | 26.00              | 28.90     | 54.00          | 60.00     |
| 10        | 2               | QPSK            | 3/4         | 39.00              | 43.30     | 81.00          | 90.00     |
| 11        | 2               | 16-QAM          | 1/2         | 52.00              | 57.80     | 108.00         | 120.00    |
| 12        | 2               | 16-QAM          | 3/4         | 78.00              | 86.70     | 162.00         | 180.00    |
| 13        | 2               | 64-QAM          | 2/3         | 104.00             | 115.60    | 216.00         | 240.00    |
| 14        | 2               | 64-QAM          | 3/4         | 117.00             | 130.00    | 243.00         | 270.00    |
| 15        | 2               | 64-QAM          | 5/6         | 130.00             | 144.40    | 270.00         | 300.00    |
| 16        | 3               | BPSK            | 1/2         | 19.50              | 21.70     | 40.50          | 45.00     |
| 17        | 3               | QPSK            | 1/2         | 39.00              | 43.30     | 81.00          | 90.00     |
| 18        | 3               | QPSK            | 3/4         | 58.50              | 65.00     | 121.50         | 135.00    |
| 19        | 3               | 16-QAM          | 1/2         | 78.00              | 86.70     | 162.00         | 180.00    |
| 20        | 3               | 16-QAM          | 3/4         | 117.00             | 130.00    | 243.00         | 270.00    |
| 21        | 3               | 64-QAM          | 2/3         | 156.00             | 173.30    | 324.00         | 360.00    |
| 22        | 3               | 64-QAM          | 3/4         | 175.50             | 195.00    | 364.50         | 405.00    |
| 23        | 3               | 64-QAM          | 5/6         | 195.00             | 216.70    | 405.00         | 450.00    |
| 24        | 4               | BPSK            | 1/2         | 26.00              | 28.80     | 54.00          | 60.00     |
| 25        | 4               | QPSK            | 1/2         | 52.00              | 57.60     | 108.00         | 120.00    |
| 26        | 4               | QPSK            | 3/4         | 78.00              | 86.80     | 162.00         | 180.00    |
| 27        | 4               | 16-QAM          | 1/2         | 104.00             | 115.60    | 216.00         | 240.00    |
| 28        | 4               | 16-QAM          | 3/4         | 156.00             | 173.20    | 324.00         | 360.00    |
| 29        | 4               | 64-QAM          | 2/3         | 208.00             | 231.20    | 432.00         | 480.00    |
| 30        | 4               | 64-QAM          | 3/4         | 234.00             | 260.00    | 486.00         | 540.00    |
| 31        | 4               | 64-QAM          | 5/6         | 260.00             | 288.80    | 540.00         | 600.00    |

# Signal constellations

QPSK



- **Symbol constellations from a 3 x 3 example**

$A1, 2, 3$  as transmitted by three TX antennas

$B1, 2, 3$  as received by three RX antennas

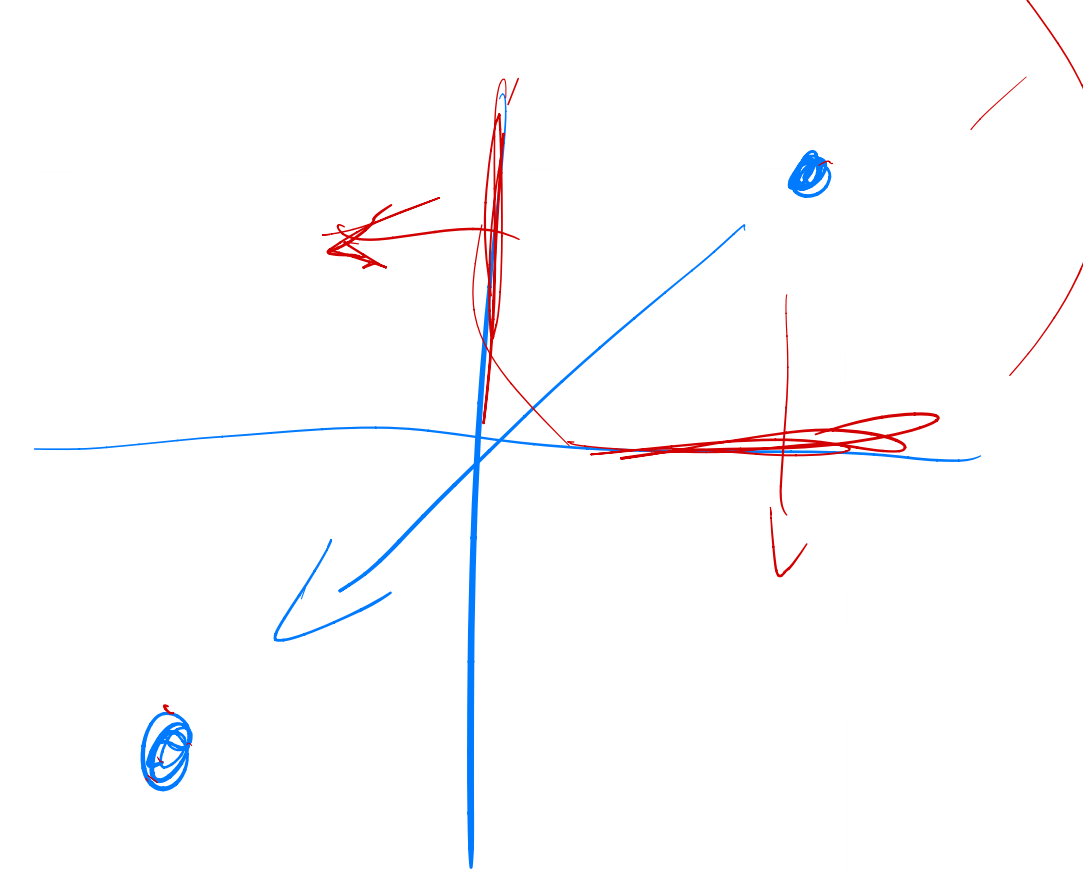
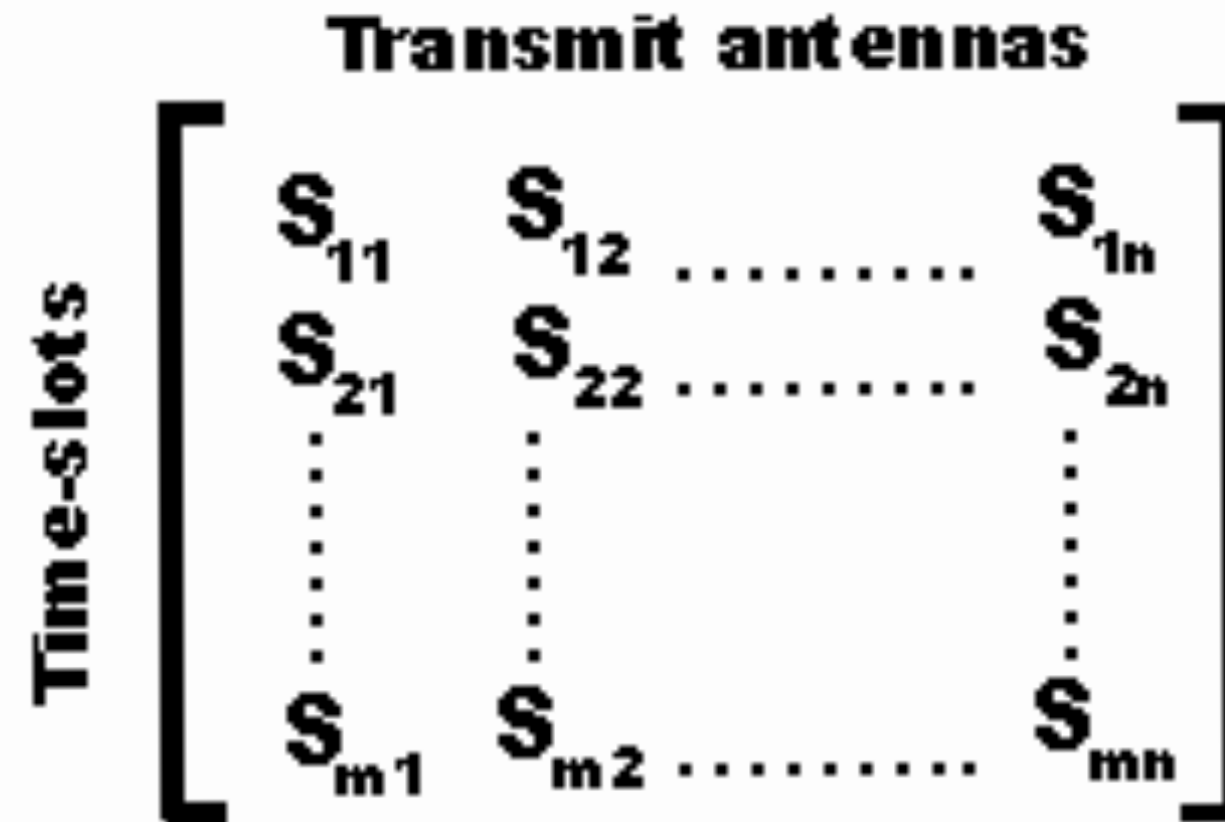
$C1, 2, 3$  after processing, at the inputs to three demodulators.

The three parallel symbol streams were derived from a single stream at 3 times the symbol rate, and are

subsequently reassembled in the original time sequence



# Space Time Block Coding



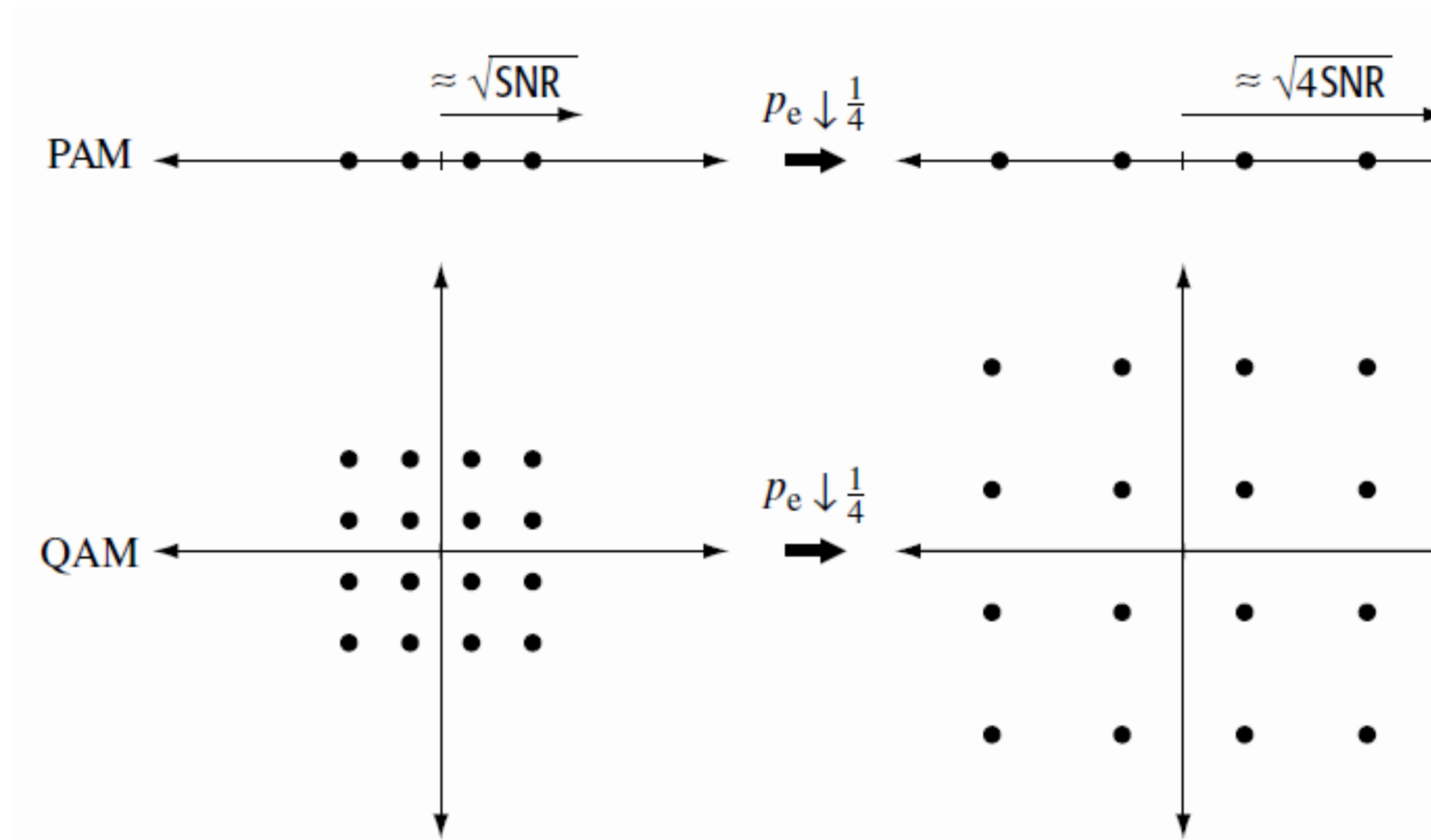
## MIMO Alamouti coding

A particularly elegant scheme for MIMO coding was developed by Alamouti.

The associated codes are often called MIMO Alamouti codes or just Alamouti codes.

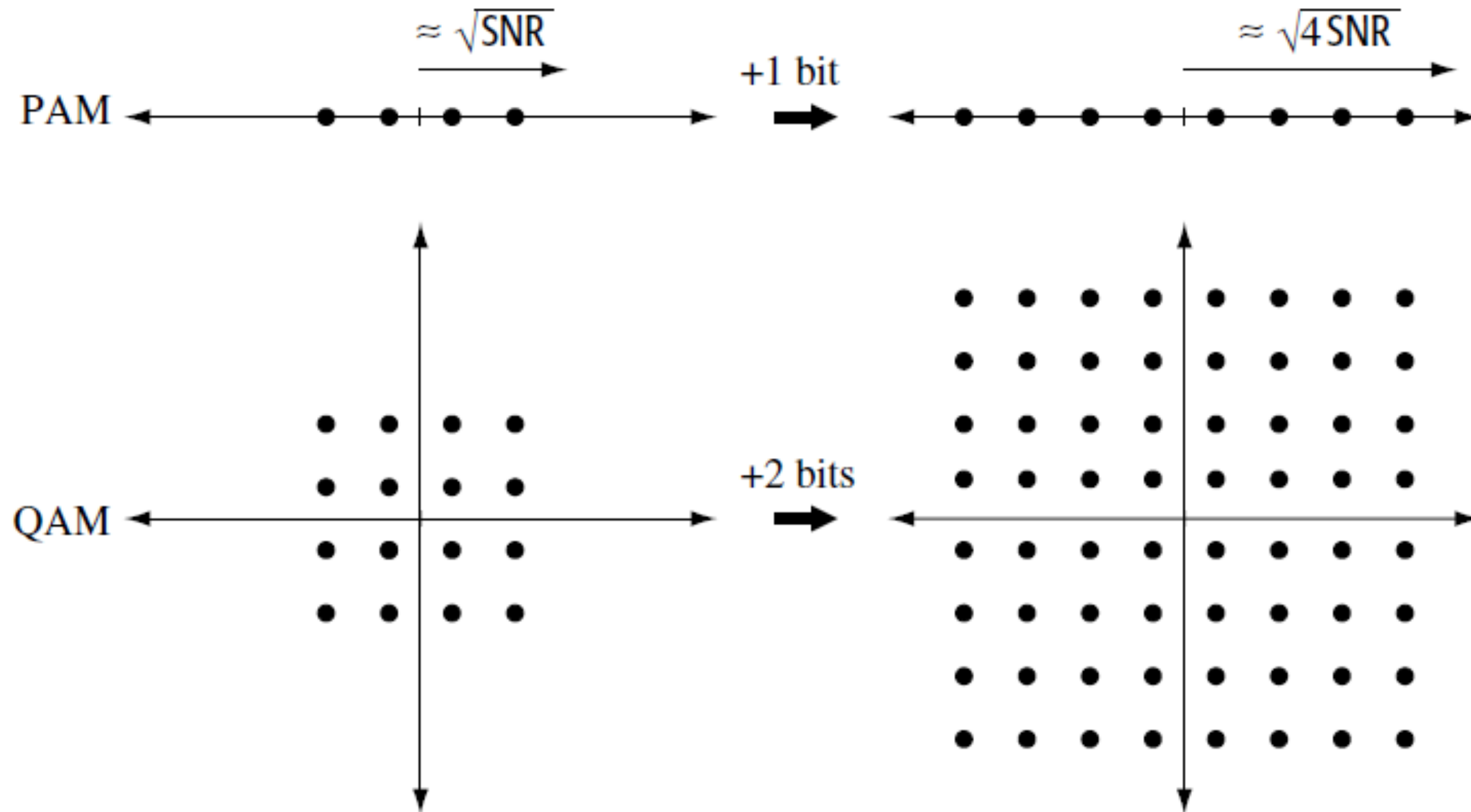


# Fixed Rate



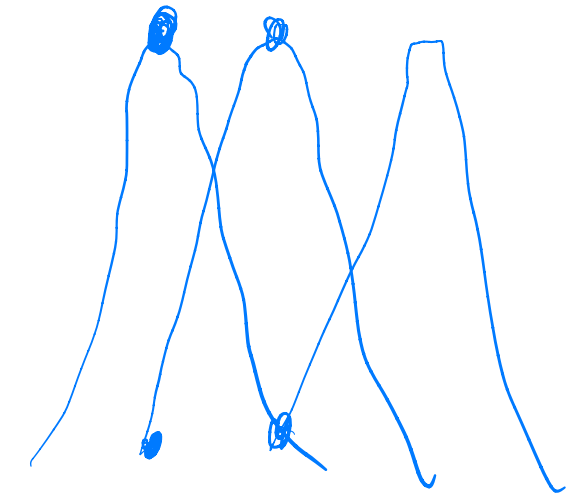


# Fixed Reliability



# High Throughput WiFi - 802.11n

a/c

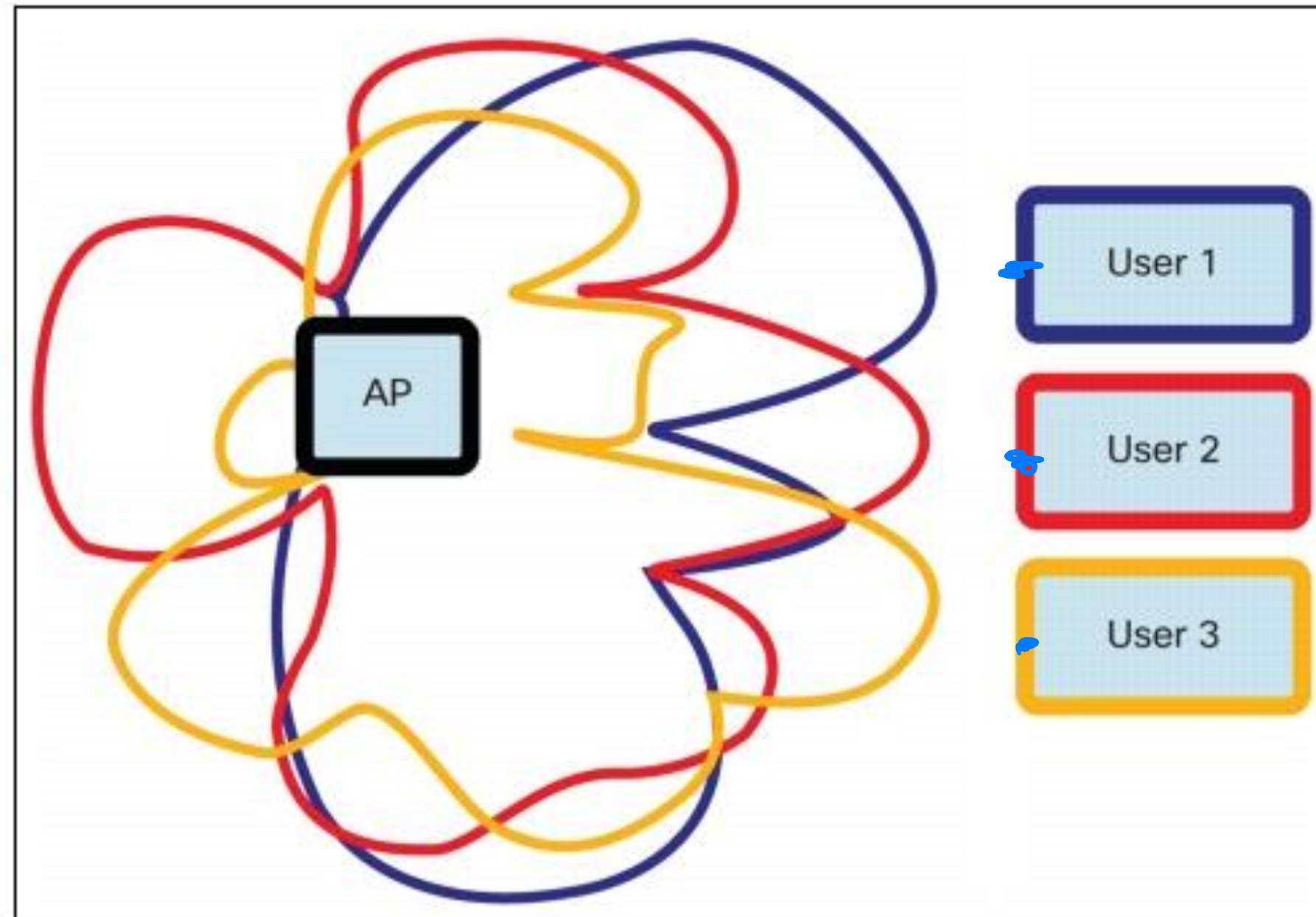


## Technology Overview

- 2.4 GHz and 5.8 GHz unlicensed bands
- Channel bandwidth of 20 MHz and 40 MHz
- Up to 4 spatial streams (e.g., 4x4)
- Current product offerings (pre-N) use only 2 spatial streams with 3Tx / 3Rx in the AP and 2Tx / 3Rx in the mobile supporting up to 300 Mbps
- Spatial diversity, spatial multiplexing, beamforming
- Enhancements in both PHY and MAC (e.g., frame aggregation, block-ACK, space-time coding, power save, green field mode, etc.)

# Multi-user MIMO

- MIMO with existing techniques



MU-MIMO uses beam-forming to spatially multiplex in the download direction

Image: Cisco

# Summary

- Looked at smart antennas
  - Gaining information, DOA
  - Using it
- Looked at receive beamforming
  - Simple (switched)
  - Clever (adaptive)
  - Very clever (SDMA)
- Looked at transmit beamforming
  - The inverse of the receive information used